

A dark, atmospheric photograph of a train station platform. A high-speed train is visible on the right, and several people are walking on the platform. Red rectangular bounding boxes are overlaid on the image, highlighting specific individuals, demonstrating the output of an AI-based person detection system. The text is centered over the image.

AI-BASED MISSING PERSON DETECTION

Smart Surveillance · Facial Recognition · Computer Vision

| THE REAL-WORLD CRISIS

2,130

People go missing daily in India.

That translates to roughly 88 people per hour. Current search methods involve manual CCTV reviews—a slow, expensive, and error-prone process that costs lives.



| CRITICAL BOTTLENECKS



TIME CONSUMPTION

Manually scanning thousands of hours of footage takes days, while the first 48 hours are critical.



MANPOWER HEAVY

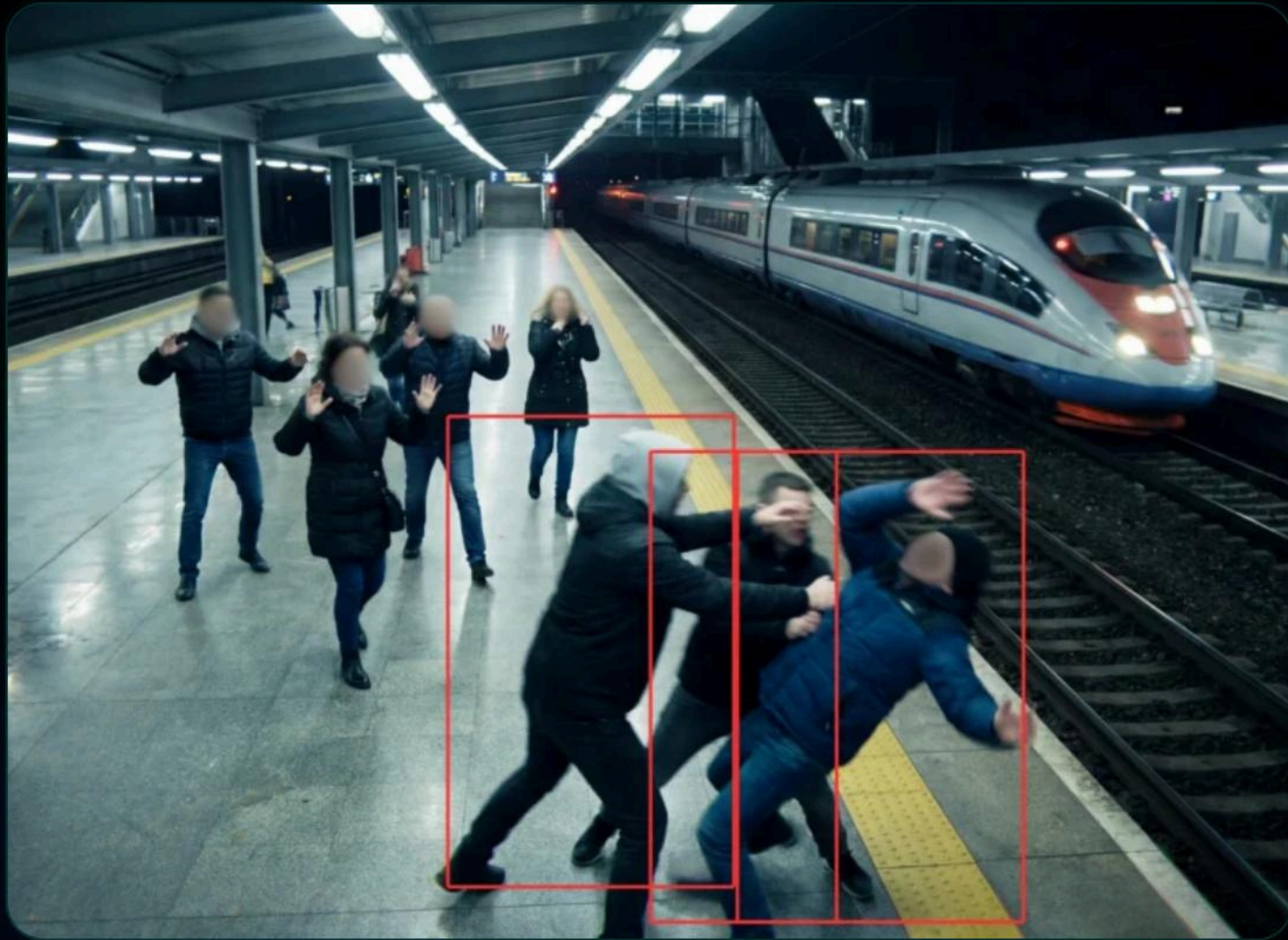
Requires constant human monitoring of hundreds of camera feeds simultaneously.



HIGH ERROR RATE

Human fatigue leads to missing subtle visual cues or faces in crowded environments.

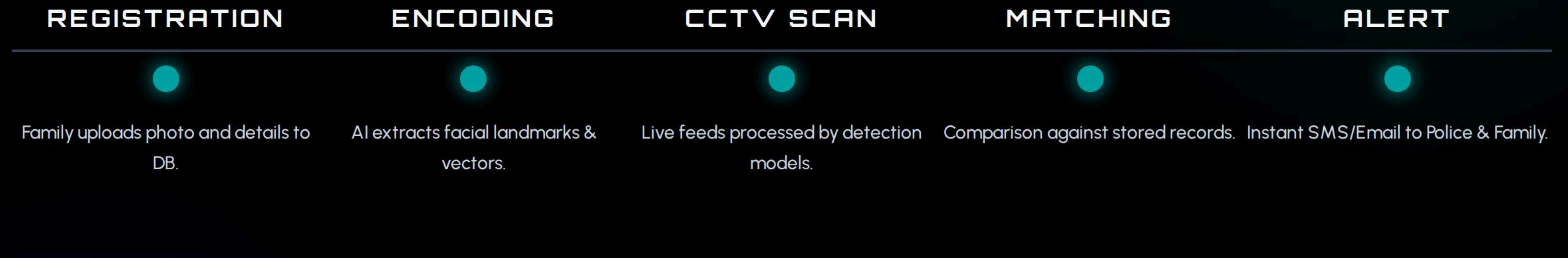
| PROPOSED SOLUTION



REAL-TIME IDENTIFICATION ENGINE

- ✓ **Automated Scanning:** Continuous monitoring of public transport hubs and malls.
- ✓ **Feature Extraction:** Converts facial landmarks into unique numerical vectors.
- ✓ **Instant Alerts:** Notifies police within seconds of a high-confidence match.
- ✓ **Scalability:** Connects thousands of cameras to a central intelligence hub.

| HOW THE SYSTEM WORKS



The system utilizes **Face Matching Similarity Scores** (e.g., Threshold > 90%) to minimize false positives.

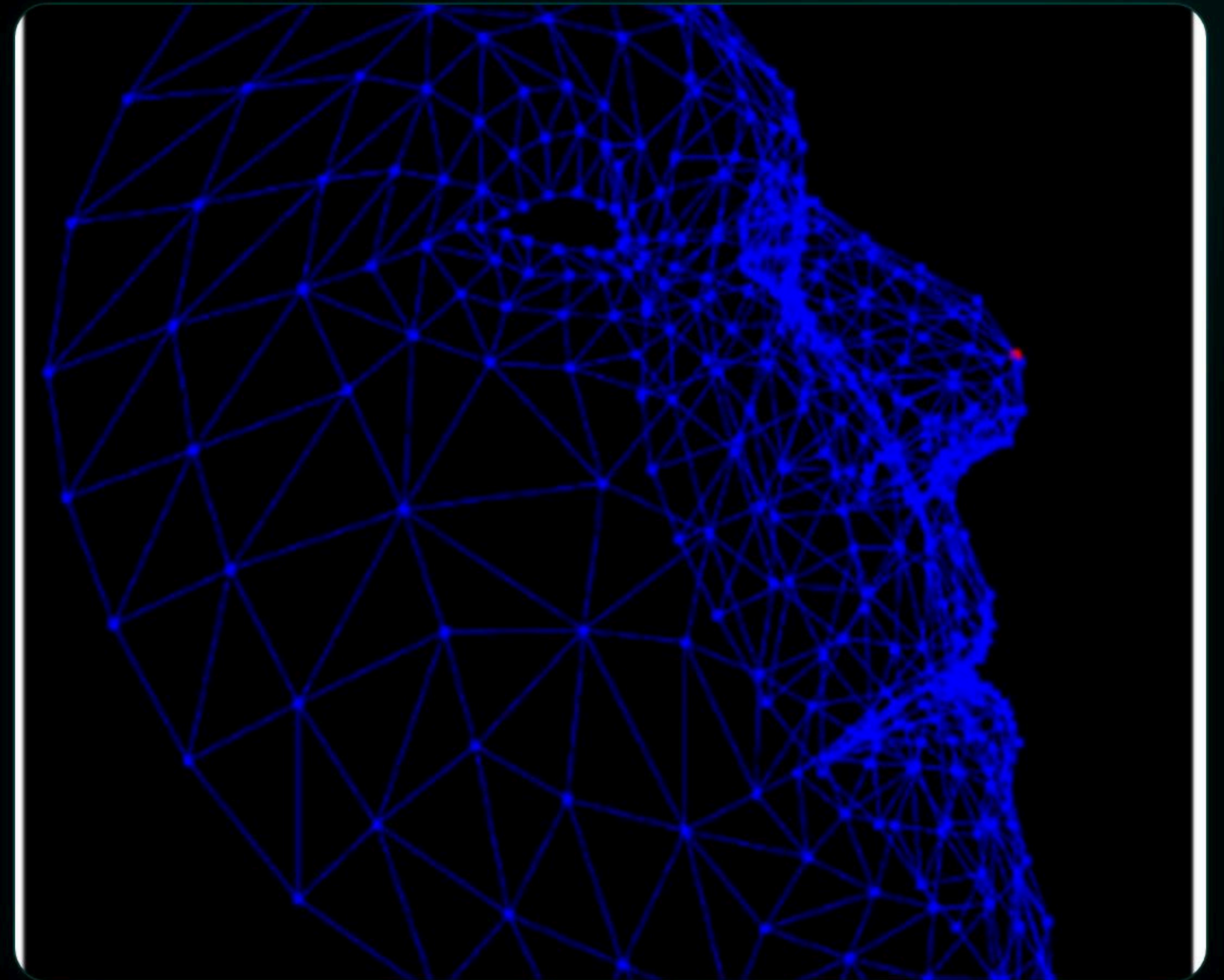
| FACE ENCODING TECHNOLOGY

MATHEMATICAL IDENTITY

Instead of raw pixels, the system stores **128 unique measurements** of the face:

- ✓ Distance between eyes and nose.
- ✓ Jawline curvature and forehead width.
- ✓ Structure of facial landmarks (68-point mesh).

The encoding works even if the person ages slightly or wears accessories.



| STRATEGIC DEPLOYMENT



TRANSPORT HUBS

Railway stations, airports, and bus terminals are primary scanning zones.



PUBLIC HUBS

Malls, shopping streets, and high-footfall parks.



SMART CITY GRID

Integration with existing municipal traffic and security cameras.



SAFE HAVENS

Surveillance near shelters and NGO centers to identify rescues.

| AUTOMATED ALERTS

When a match is found, the system generates a priority notification containing:

- ✓ **Location:** GPS & Camera ID.
- ✓ **Timestamp:** Exact second of detection.
- ✓ **Accuracy:** Confidence percentage (e.g., 96%).

"Alert sent to Coimbatore Railway Police. Confidence 97%."

| THE TECHNOLOGY STACK



- ✓ **Python:** Core logic and system integration.
- ✓ **OpenCV:** Real-time image and video stream processing.
- ✓ **Deep Learning:** DLib & Face_Recognition models for high accuracy.
- ✓ **YOLO:** Efficient multi-person detection in frames.
- ✓ **PostgreSQL:** Scalable database for secure records.

| WHY IT WORKS



FASTER SEARCH

Weeks of search reduced to seconds.



24/7 MONITORING

Works tirelessly without fatigue.



COST EFFECTIVE

Significant reduction in investigative overhead.

Success Case: 3,000 children found in 4 days in New Delhi using similar AI tech.

| FUTURE ENHANCEMENTS



AGE PROGRESSION

AI to predict current appearance for long-term missing cases.



DRONE INTEGRATION

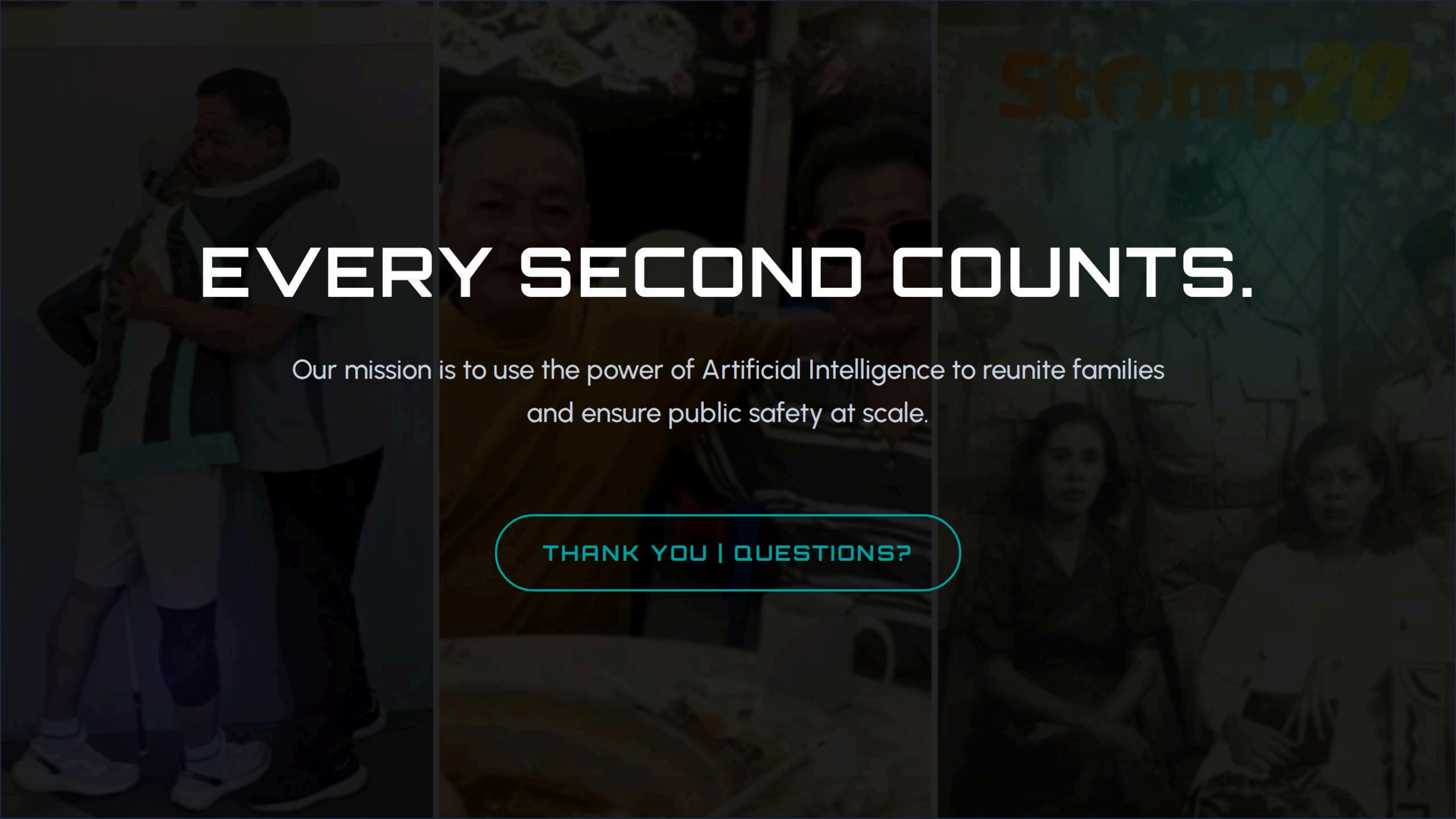
Search large disaster zones and remote areas using aerial AI.



VOICE RECOGNITION

Multi-modal identification using voice sample matching.

Aim: A nationwide interconnected database linking all police stations for unified tracking.

The background is a collage of three images. The left image shows a man with a cane and a woman. The middle image shows a family sitting at a table. The right image shows a group of people, with a 'Stamp20' sign visible in the top right corner.

EVERY SECOND COUNTS.

Our mission is to use the power of Artificial Intelligence to reunite families
and ensure public safety at scale.

THANK YOU | QUESTIONS?